

Dahyun Joo

+82-10-2647-9797 | clairda@snu.ac.kr | springfall137.github.io

Republic of Korea | DOB: Aug 28, 2000

ACADEMIC BACKGROUND

Seoul National University <i>M.S. Course, Mechanical Engineering, GPA 4.0/4.0</i> AI-driven Simulation and Design Lab (Principal Investigator: Prof. Do-Nyun Kim)	2025 – Present
Seoul National University <i>B.S. Physics (Minor: Mechanical Engineering), Cum Laude</i> Thesis: <i>wPINNs: Enhancing Physics-Informed Neural Networks with Wavelet Activation Functions</i>	2019 – 2025
Chungnam Science High School <i>R&E at Physics Student Group, Valedictorian</i>	2016 – 2019

RESEARCH EXPERIENCE

M.S. Researcher <i>SNU AI-driven Simulation and Design Lab (Prof. Do-Nyun Kim)</i> - Develop reusable kirigami-based impact absorbers by engineering geometry-driven negative-to-positive stiffness transitions and energy absorption for repeatable deformation without material damage. - Build simulation-to-design pipelines (e.g., FEA, parametric modeling, optimization loops) to map pattern parameters to force–displacement response and impact performance. - Develop an explainable ten-qubit quantum neural network for multi-material topology optimization, with transfer to unseen loading conditions, higher-resolution meshes, and voxel-wise 3D problems.	Mar 2025 – Present
Undergraduate Research Intern <i>SNU MetaStructure Lab (Prof. J. H. Oh)</i> Studied waves and metamaterials using acoustic finite element analysis and supported AI-driven topology-optimization simulations and design iterations.	Sep 2024 – Jan 2025
Undergraduate Research Intern <i>SNU Transformative Architecture Laboratory (Prof. J. K. Yang)</i> Performed Abaqus FEA of deployable bistable dodecahedral origami to assess fabrication feasibility and guide design iteration.	Oct 2023 – Jan 2024

PUBLICATIONS

D. Joo, Y. Miyazawa, S. Hong, D. Lee, J. K. Yang, D.-N. Kim, <i>Kirigami Meta-Sheet for Reusable Impact Absorption</i>. Manuscript in preparation, 2026. Introduces a planar, reusable kirigami impact absorber that dissipates local impacts through distributed deformation and transitions between positive- and negative-stiffness regimes, validated through experiments, finite-element simulations, and reduced-order modeling.	
D. Joo, N. Sukulthanasorn, K. Terada, D.-N. Kim, <i>Explainable Quantum Neural Networks for Multi-Material Topology Optimization</i>. Submitted to <i>Nature Machine Intelligence</i>; arXiv:2607.00438 [cs.CE], 2026. Develops a ten-qubit explainable quantum neural network that predicts structural layouts and material assignments from multi-material topology-optimization histories, generalizing to unseen loading conditions, higher-resolution meshes, and voxel-wise 3D problems while retaining auditable observable channels.	

PROJECTS

Samsung Display <i>AI-driven Pattern Discovery for Improved Hinge Deformation Behavior</i> Built an automated 2D modeling–simulation workflow over unit-cell and hinge geometries, generating AI-training datasets and a surrogate model for hinge-strain prediction and geometry selection.	2026.03 – 2027.02
Samsung Display <i>Bistable Kirigami Integration for Curved Display Corner</i> Integrated and parameterized bistable kirigami patterns for controlled shape transition under corner-curvature, boundary, manufacturability, and repeatability constraints.	2025.03 – 2026.02

Kirigami-Enhanced Pulp-Mold Packaging for Impact Absorption

- Developed kirigami-enhanced pulp-mold packaging concepts and practical pattern guidelines for improved impact absorption under large-scale forming constraints.

PATENTS

Samsung Display, D.-N. Kim, **D. Joo**, G. Yun, K. Jung, *Bistable Metapatterns with Dome Geometry and Optimal Design Method*, patent pending.

AWARDS

STOB League: Semiconductor Solverton Competition, Top Prize	2024
• Deputy Prime Minister and Education Minister Prize. • Optimized GAA nanosheet FET AC performance using structural modeling and TCAD; analyzed resistance, capacitance, and frequency gain relative to FinFET.	
Seoul National University Semiconductor Specialized College Scholarship	2024
Korean History Proficiency Test (Level 1, 100/100)	2022
ROKAF Entrepreneurship Contest: Advanced to MND Finals	2022
Excellent Private Award	2021
• Granted by the ROK Air Force Information Communication School Principal.	
Certificate of Completion: Startup Investment Education	2020
• Issued by the SNU Entrepreneurship Center.	
Excellent Student Award	2019
• Granted by the Chungcheongnam-do Federation of Teachers' Association.	

SEMINAR

Seminar Instructor (Reinforcement Learning)	Jan 2024
<i>SNU BI Lab (Prof. Byoung-Tak Zhang)</i>	
• Translated reinforcement-learning concepts into accessible frameworks for an interdisciplinary audience. • Explained game-theoretic foundations of modern RL, including minimax decision-making, Nash equilibrium, and links to policy/value learning.	

PROFESSIONAL EXPERIENCE

Intern — K-Startup Center Paris at STATION F	2025.02
Intern — Thales Korea	2024.03 – 2024.07
Working Scholar — SNU Gwanak Residence Halls Administrative Office	2024.01 – 2024.02
Content Creator — @biztorang & @jjuvoyager	2022.10 – 2023.08
Sergeant (Completed Service) — Republic of Korea Air Force (15th Weather Squadron)	2021.07 – 2023.04
• Developed leadership and operational-coordination skills in a high-responsibility environment.	
President of Student Council — SNU Department of Physics and Astronomy	2020.11 – 2021.07
• Led student representation and cross-group coordination.	
Working Scholar — Center for Theoretical Physics, SNU Department of Physics and Astronomy	2020.03 – 2021.06
Brand Owner / Online Retail Sales — jjooda & DogodoNongwon	2020.04 – 2023.05
CFD Engineer — Rocket pre-startup Kspace (Injector Design)	2019.12 – 2020.02

SKILLS

Languages: Korean (Native), English (TOEFL 104; GRE V155/Q170/AW4.0), French (DELFI A2)
Programming: C, Python, MATLAB, R | **Engineering:** ABAQUS, ANSYS, NX, STAR-CCM+, SolidWorks, COMSOL, KeyShot, Cura
Fabrication / Testing: FDM, SLA and metal 3D printing; laser cutting; UTM; milling